# 1ª Parte – Teoría de curvas

## Producto escalar

Forma $\begin{cases}\text{bilineal}\\ \text{simétrica}\\ \text{def. positiva}\end{cases}$

↳ Base $\mathcal{B}=\{\vec{e}_1,\dots,\vec{e}_n\}$: ortogonal $\vec{e}_i\cdot\vec{e}_j=0,\ i\neq j$; ortonormal $e_i\cdot e_j=0,\ i\neq j$ ; $e_i\cdot e_i=1$

## Desigualdades: $(\vec{V},\cdot)$ E.V. euclídeo; $a,b\in\vec{V}$

- Norma: $\|\vec{a}\|=\sqrt{\vec{a}\cdot\vec{a}}$
- Ángulo: $\alpha=\arccos\frac{\vec{a}\cdot\vec{b}}{\|\vec{a}\|\cdot\|\vec{b}\|}\in[0,\pi]$
  ↳ $\vec{a}\cdot\vec{b}=\|\vec{a}\|\cdot\|\vec{b}\|\cdot\cos\alpha$
- Vector unitario: $\vec{u}_{\vec{a}}=\frac{\vec{a}}{\|\vec{a}\|}\Rightarrow\|\vec{u}_{\vec{a}}\|=1$
- Proyección ortogonal: $\vec{p}_{\vec{a}}(\vec{v})=\frac{\vec{a}\cdot\vec{b}}{\|\vec{b}\|}\vec{u}_{\vec{b}}\Rightarrow\cos\alpha=\frac{\|\vec{p}_{\vec{a}}(\vec{a})\|}{\|\vec{a}\|}$
- Producto escalar Standard $\vec{a}\cdot\vec{b}=a_1b_1+a_2b_2+a_3b_3$ $\forall\vec{a},\vec{b}\in\vec{V}$

## Producto vectorial

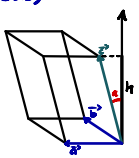
$$\vec{a}\times\vec{b}=\begin{vmatrix}\vec{e}_1&\vec{e}_2&\vec{e}_3\\ a_1&a_2&a_3\\ b_1&b_2&b_3\end{vmatrix}$$

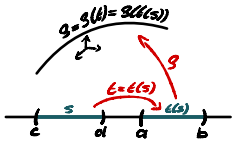
### Propiedades

- $\|\vec{a}\times\vec{b}\|=\|\vec{a}\|^2\|\vec{b}\|^2-(\vec{a}\cdot\vec{b})^2$
- $\|\vec{a}\times\vec{b}\|=\|\vec{a}\|\,\|\vec{b}\|\operatorname{sen}\alpha$
- $\vec{a}\times\vec{b}\perp\vec{a}$ y $\vec{a}\times\vec{b}\perp\vec{b}$ $\qquad\vec{a}\perp\vec{b}\Leftrightarrow\vec{a}\cdot\vec{b}=0$
- $\vec{a}\times\vec{b}=\vec{0}\Leftrightarrow\vec{a},\vec{b}$ son LD
- $\vec{a}\times\vec{b}\neq\vec{0}\Rightarrow\{\vec{a},\vec{b},\vec{a}\times\vec{b}\}$ base ⊕ orientada (c.s.m. LI)
- $\vec{a}\times\vec{b}=-\vec{b}\times\vec{a}$
- $(\lambda\vec{a}+\mu\vec{b})\times\vec{c}=\lambda\vec{a}\times\vec{c}+\mu\vec{b}\times\vec{c}$

## Producto triple (0 si nulo) (Volumen paralelepípedo)

$$[\vec{a},\vec{b},\vec{c}]=\vec{a}\cdot(\vec{b}\times\vec{c})=(\vec{a}\times\vec{b})\cdot\vec{c}=\begin{vmatrix}a_1&a_2&a_3\\ b_1&b_2&b_3\\ c_1&c_2&c_3\end{vmatrix}$$




## Regla de la cadena

$$(J_{g\circ t})_{a}=(J_g)_{t(a)}\,(J_t)_a$$

$$\left(\frac{dg}{ds}\right)_{s=a}=\left(\frac{dg}{dt}\right)_{t=t(a)}\left(\frac{dt}{ds}\right)_{s=a}$$

## Param. regular: $\begin{cases}g\text{ es de clase }C^1\\ \vec{g}'(t)\neq\vec{0}\ \ \forall t\end{cases}$

## Param. natural: $\begin{cases}g=g(s)\text{ regular}\\ \|g'(s)\|=1\end{cases}$

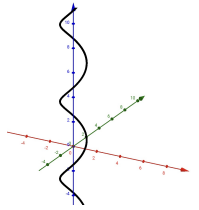
## Longitud de arco

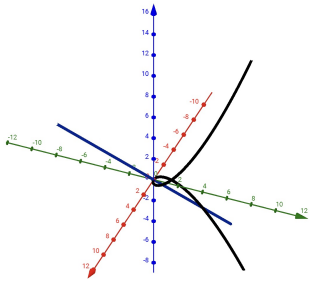
$$\int_{t_1}^{t_2}\|g'(t)\|\,dt$$

Regular $\xrightarrow{s=\int_{t_0}^{t}\|g'(u)\|\,du}$ Natural

| ‡: Depende del cambio de parametrización | $g=g(s)$ natural | $g=g(t)$ cualquiera |
|---|---|---|
| $\vec{v}$ tangente ‡ | $\vec{t}(s)=g'(t)$ | $\vec{t}(t)=\frac{g'(t)}{\|g'(t)\|}$ |
| $\vec{v}$ normal | $\vec{n}(s)=\frac{g''(s)}{\|g''(s)\|}$ | $\vec{n}(t)=\vec{b}(t)\times\vec{t}(t)$ |
| $\vec{v}$ binormal ‡ | $\vec{b}(s)=\vec{t}(s)\times\vec{n}(s)$ | $\vec{b}(t)=\frac{g'(t)\times g''(t)}{\|g'(t)\times g''(t)\|}$ |
| $\vec{v}$ curvatura | $\vec{k}(s)=g''(s)$ | $\vec{k}(t)=k(t)\,\vec{n}(t)$ |
| Curvatura | $k(s)=\|g''(s)\|$ | $k(t)=\frac{\|g'(t)\times g''(t)\|}{\|g'(t)\|^3}$ |
| Torsión | $\tau(t)=\frac{[g'(s),g''(s),g'''(s)]}{\|g''(s)\|^2}$ | $\tau(t)=\frac{[g'(t),g''(t),g'''(t)]}{\|g'(t)\times g''(t)\|^2}$ |

## Hélices generales $\exists v\in\mathbb{R}^3:\ \vec{t}\cdot\vec{v}=\cos\alpha$ cte (Eje de la hélice)





Tª Lancret: $\frac{\tau(s)}{k(s)}=\text{cte}=\cot g\,\alpha$

$\vec{v}=\frac{\tau(s)}{k(s)}\vec{t}(s)+\vec{b}(s)$

$[g''(s),g'''(s),g^{iv}(s)]=0\ \ \forall s$

## Curvas de Bertrand

Si admite una conjugada de Bertrand distinta de sí misma

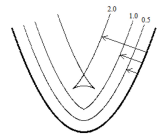
$\exists A,B\in\mathbb{R},\ A\neq0$ tq $Ak(t)+B\tau(t)=1$

## Conjugadas de Bertrand → $g$ y $\bar{g}$ son conjugadas de Bertrand si ∃ homeomorfismo entre sus puntos tq puntos correspondientes tengan la misma recta normal

$g=g(s)$ natural $\Rightarrow\bar{g}(s)=g(s)+a(s)\vec{n}_g(s)$

- Distancia entre pts correspondientes es cte
- Tangentes a la curva mantienen ángulo cte

$\vec{n}_{\bar{g}}=\pm\vec{n}_g$



## Circunferencia osculatriz

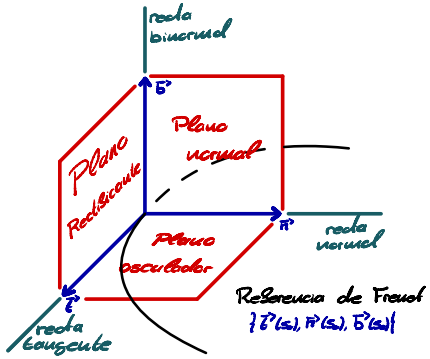
Radio: $\rho(s)=\frac{1}{k(s)}$

Centro: $c(s)=g(s)+\rho(s)\vec{n}(s)=(x_0,y_0)$

Ec. circunferencia: $\{(x-x_0)^2+(y-y_0)^2=\rho^2(s)\}$

## Fórmulas de Frenet

$$\begin{pmatrix}\vec{t}'(t)\\ \vec{n}'(t)\\ \vec{b}'(t)\end{pmatrix}=\begin{pmatrix}0&k&0\\ -k&0&\tau\\ 0&-\tau&0\end{pmatrix}\begin{pmatrix}\vec{t}(t)\\ \vec{n}(t)\\ \vec{b}(t)\end{pmatrix}$$




Referencia de Frenet $\{\vec{t}(s),\vec{n}(s),\vec{b}(s)\}$

## Evoluta y evolventes

$g$ evoluta de $\bar{g}$ ; $\bar{g}$ evolvente de $g$ si $\begin{cases}g\in\text{Sup. tangencial de }\bar{g}\\ \bar{g}\text{ corta }\perp\text{ a las rectas tangentes a }g\end{cases}$

## Superficie tangencial: $F(s,\lambda)=g(s)+\lambda\vec{t}_g(s)$

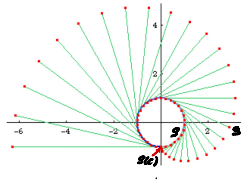
$g=g(s)$ natural $\Rightarrow\bar{g}(s)=g(s)+(c-s)\vec{t}_g(s)$

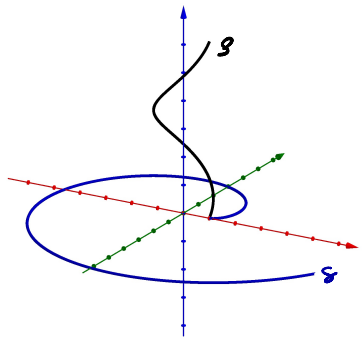
$\|\bar{g}(s)=g(s)+\lambda(s)\vec{t}_g(s)\ \|\ g'(s)=\lambda\overrightarrow{\bar{g}(s)g(s)}$

$g'(s)\cdot\bar{g}'(s)=0$

$g(c)=\bar{g}(c)$

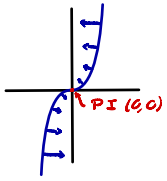
$\int_c^{s}\|g'(s)\|\,ds=\|\overrightarrow{g(s_0)\bar{g}(s_0)}\|$

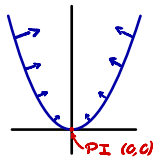



## Punto de inflexión

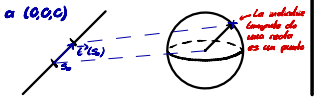
$g(s_0)$ es PI si $k(s_0)=0$



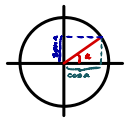

## Indicatrices $\begin{cases}\text{tangente}\\ \text{normal}\leftarrow(\text{análoga})\\ \text{binormal}\end{cases}$

Indicatriz $\vec{t}$ define una curva en $S^2=\{x^2+y^2+z^2=1\}$ trasladando el vector $\vec{t}$ en cada punto de la curva a $(0,0,0)$

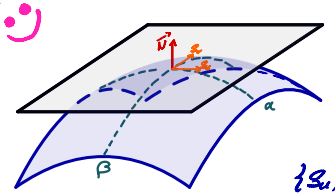



## Recta en $(x_0,y_0)$ y vector $\vec{v}(x,y)$

$(x,y,z)=g(x_0,y_0)+\lambda v(x_0,y_0)$




| | 0 | $\pi/6$ | $\pi/4$ | $\pi/3$ | $\pi/2$ |
|---|---|---|---|---|---|
| sen | 0 | $1/2$ | $\sqrt{2}/2$ | $\sqrt{3}/2$ | 1 |
| cos | 1 | $\sqrt{3}/2$ | $\sqrt{2}/2$ | $1/2$ | 0 |




$$\vec{N}(u_0,v_0) = \frac{g_u(u_0,v_0)\times g_v(u_0,v_0)}{\|g_u(u_0,v_0)\times g_v(u_0,v_0)\|}$$

$\|\vec{N}\| = 1$; $\vec{N} \perp g_u$; $\vec{N} \perp g_v$

$\{g_u, g_v, \vec{N}\}$ base positivamente orientada

## Líneas Coordenadas

$\alpha(t) = g(u_0+t, v_0) \quad t \in (-\varepsilon, \varepsilon)$

$\beta(t) = g(u_0, v_0+t) \quad t \in (-\varepsilon, \varepsilon)$

## Superficie Simple

1. $g_u \times g_v = \vec{0}$ $\forall$ punto de $U$
2. $g: U \to g(U)$ homeomorfismo $\begin{cases} g \text{ biyectiva} \\ g \text{ continua} \\ g^{-1} \text{ continua} \end{cases}$

↳ Ej. Sup de Monge: $g(u,v) = g(u,v,h(u,v))$

## Superficie Regular

① $\forall p \in S$ $\exists U \subseteq \mathbb{R}^2$ abierto, $g: U \to g(U) \in \mathbb{R}^3$ Sup. simple, $p \in g(U) \in \mathbb{R}^3$

② $g: U \to V \subseteq S$, $\bar{g}: \bar{U} \to \bar{V} \subseteq S$ sup. simples, $V \cap \bar{V} = \emptyset \Rightarrow$ $\Rightarrow \bar{g}^{-1} \circ g$ difeomorfismo

Sup simple $\Rightarrow$ Sup regular $\Rightarrow$ Sup topológica

$S = F^{-1}(0)$, $(\nabla F)_p \neq (0,0,0)$ $\forall p = (p_1,p_2,p_3) \in F^{-1}(0)$

$\Rightarrow \begin{cases} S \text{ sup regular} \\ T_p S = \{F_x(p)(x-p_1) + F_y(p)(y-p_2) + F_z(p)(z-p_3) = 0\} \end{cases}$

## Primera Forma Fundamental

$$g = \begin{pmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{pmatrix} \qquad g^{-1} = \begin{pmatrix} g^{11} & g^{12} \\ g^{21} & g^{22} \end{pmatrix} = \frac{1}{\det g}\begin{pmatrix} g_{22} & -g_{12} \\ -g_{21} & g_{11} \end{pmatrix}$$

$g_{ij} = g_{u_i} \cdot g_{u_j}$ $\begin{pmatrix} u_1 = u \\ u_2 = v \end{pmatrix}$

## Segunda Forma Fundamental

$$L = \begin{pmatrix} L_{11} & L_{12} \\ L_{21} & L_{22} \end{pmatrix} \qquad L_{ij} = g_{u_i u_j} \vec{N}$$

## Ec. de Gauss

$$g_{u_i u_j} = \sum_{k=1}^{2} \Gamma_{ij}^{k} g_{u_k} + L_{ij} \vec{N}$$

$\Gamma_{ij}^{k}$: Símbolos de Christoffel

## Ec. de Weingarten

$$\vec{N}_{u_i} = \sum_{k=1}^{2} \beta_i^{\,k} g_{u_k}$$

$$\beta_1^{\,1} = \frac{-L_{11}g_{22} + L_{21}g_{21}}{\det g} \qquad \beta_1^{\,2} = \frac{-L_{21}g_{11} + L_{11}g_{12}}{\det g}$$

$$\beta_2^{\,1} = \frac{-L_{12}g_{22} + L_{22}g_{12}}{\det g} \qquad \beta_2^{\,2} = \frac{-L_{22}g_{11} + L_{12}g_{12}}{\det g}$$

## Aplicación de Weingarten

- Respecto de $\{g_u, g_v\}$

$$\mathcal{L} = g^{-1}L = \begin{pmatrix} -\beta_1^{\,1} & -\beta_2^{\,1} \\ -\beta_1^{\,2} & -\beta_2^{\,2} \end{pmatrix}$$

$\vec{N}_u = -\mathcal{L}(g_u)$

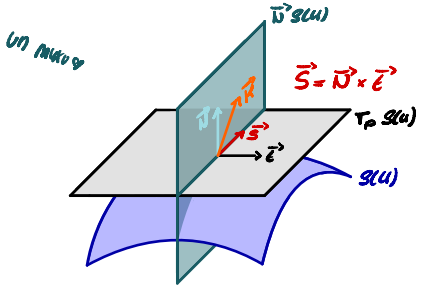
$\vec{N}_v = -\mathcal{L}(g_v)$

- Respecto de $\{\vec{e}_1, \vec{e}_2\}$ (vectores propios (*))

$$\mathcal{L} = \begin{pmatrix} k_1 & 0 \\ 0 & k_2 \end{pmatrix} \Rightarrow \begin{cases} \mathcal{L}(\vec{e}_1) = k_1\vec{e}_1 \\ \mathcal{L}(\vec{e}_2) = k_2\vec{e}_2 \end{cases}$$

($k_1, k_2$: valores propios)

- $L(x,y) = g(\mathcal{L}(x), y) = g(x, \mathcal{L}(y))$




Curvatura normal: $\vec{k} = k_n \vec{N} + k_g \vec{S}$ (Curvatura geodésica)

$k_n = \mathcal{L}(\vec{v}, \vec{v}) = \frac{L(\vec{v},\vec{v})}{g(\vec{v},\vec{v})}$, $\vec{v}$ tangente

## Ec. de Weingarten

$$\vec{N}_{u_i} = \sum_{k=1}^{2} \beta_i^{\,k} g_{u_k}$$

$$\beta_1^{\,1} = \frac{-L_{11}g_{22} + L_{21}g_{21}}{\det g} \qquad \beta_1^{\,2} = \frac{-L_{21}g_{11} + L_{11}g_{12}}{\det g}$$

$$\beta_2^{\,1} = \frac{-L_{12}g_{22} + L_{22}g_{12}}{\det g} \qquad \beta_2^{\,2} = \frac{-L_{22}g_{11} + L_{12}g_{12}}{\det g}$$

## Curvaturas ppales

valores propios $k_1$ y $k_2$ ← pto umbílico $k_1 = k_2$

Curvaturas ppales $k_1$ y $k_2$

Curvatura de Gauss $K = k_1 k_2 = \frac{\det L}{\det g} = \det \mathcal{L} \Rightarrow$

- $K < 0$ pto hiperbólico — $D_p$ ∅
- $K = 0$ < $k_1 = 0 = k_2$ pto plano; $k_1 \neq k_2$ pto parabólico — ↳ $D_p$ ∥
- $K > 0$ pto elíptico — $D_p$ ○ ∧ ∅

Curvatura media $H = \frac{1}{2}(k_1 + k_2) = \frac{1}{2}\mathrm{tr}\,\mathcal{L}$

Direcciones ppales: rectas generadas por $\vec{e}_1$ (*), $\vec{e}_2$

Línea de curvatura: tangente a una dirección ppal

Dirección asintótica: Aquellas donde $k_n = 0$

## Pol. CTCO

$\chi_{\mathcal{L}}(t) = t^2 - (k_1+k_2)t + k_1k_2 = t^2 - 2Ht + K$

## Tª Euler

$k_n(\gamma) = k_1\cos^2\theta + k_2\sin^2\theta$

$\theta = \widehat{\vec{e}_1\gamma}$, $\gamma \in T_p\, g(U)$, $|\gamma| = 1$

## Indicatriz de Dupin

$D_p = \{x \in T_p\, g(U) : L(x,x) = \pm 1\}$

$= \{x \in T_p\, g(U) : x^2(k_1+k_2) = \pm 1\}$ ($k_1x^2 + k_2x^2 = \pm 1$)

## Procedimiento

1. Calcular la 1ª forma fundamental
2. Calcular la 2ª forma fundamental
   ↳ Curvatura de Gauss $K = \frac{\det L}{\det g}$
3. Calcular la aplicación de Weingarten $\mathcal{L} = g^{-1}L$
   ↳ Curvatura de Gauss $K = \det \mathcal{L}$
   ↳ Curvatura media $H = \frac{1}{2}\mathrm{tr}(\mathcal{L})$
   ↳ Curvaturas ppales: raíces de $\chi_{\mathcal{L}}(t) = t^2 - 2Ht + K$

| Punto | Dupin | Direcciones donde $k_n = 0$ |
|---|---|---|
| Elíptico | Elipse ∧ ∅ | Ninguna |
| Hiperbólico | 2 hipérbolas conjugadas | Asíntotas de $D_p$ |
| Parabólico | 2 rectas ∥ | ∥ a rectas de $D_p$ |
| Plano | ∅ | Cualquier dirección |

## Tª Meusnier

Si $\alpha(a) = \beta(b)$ y $\vec{t}_\alpha(a) = \vec{t}_\beta(b)$

$\Rightarrow (k_n)_\alpha(a) = (k_n)_\beta(b)$

# 2ª Parte

# Teoría de Superficies